RESEARCH ARTICLE

Comparing the Influence of Conventional and Rotary Instrumentation Techniques on the Behavior of the Children: A Randomized Clinical Trial

Lavanya Govindaraju[1], EMG Subramanian[2], Ganesh Jeevanandan[3]

Introduction

Early childhood caries is a globally existing oral health condition possessing a strong risk of primary teeth getting lost at an earlier age. Early loss of primary teeth endures to be a major concern in the field of pediatric dentistry. The role of a pedodontist is to maintain the primary teeth in its position until exfoliation without signs of infection, thus preserving the integrity of the dental arches.[1] Hence, endodontic management of primary teeth with infected pulp became the growing field in pediatric dentistry.

Pulpectomy is the only choice of treating symptomatic primary teeth in children to retain the primary teeth in its natural position as a fully functional component in the dental arch. The procedure of pulpectomy involves complete pulpal removal followed by debridement of the canal space for adequate obturation with a resorbable root filling material.[2] Haapasalo et al. have reported that bio-mechanical preparation is the most important step that determines the success of the pulp therapy.[3] The primal intent of root canal preparation in primary teeth is to completely eliminate the organic debris in relatively shorter time.[4]

Hand files were used for canal debridement of the primary teeth traditionally and extensively but have also been reported to be more time consuming and result in iatrogenic errors.[5] To overcome the detriments of the hand instrumentation in primary roots canals and to preserve the torturous root canal path of the primary teeth, NiTi instrumentation was introduced into pediatric dentistry.[6] Besides the limitation in the procedural errors and reduction of the operator's fatigue, the major advantages of using rotary instruments over hand instrumentation in pediatric dentistry is the reduction in the instrumentation time and its positive influence on the child's behavior towards dental treatment. The length of the appointment is strongly associated with the child's behavior.[7,8]

The behavior of children plays an integral component and is the primary requisite of the pediatric dental practice.[9,10] Assessment of child's behavior during a dental procedure is one

[1–3]Department of Pedodontics, Saveetha Dental College, Chennai, Tamil Nadu, India

Corresponding Author: Ganesh Jeevanandan, Department of Pediatric and Preventive Dentistry, Saveetha Dental College, Chennai, Tamil Nadu, India, Phone: +91 9884293869, e-mail: helloganz@gmail.com

How to cite this article: Govindaraju L, Subramanian EMG, Jeevanandan G. Comparing the Influence of Conventional and Rotary Instrumentation Techniques on the Behavior of the Children: A Randomized Clinical Trial. Int J Clin Pediatr Dent 2021;14(S-2):S179–S185.

Source of support: Nil

Conflict of interest: None

of the most important skills of a pediatric dentist for successful completion of the dental treatment.[11] The child's behavior on a dental visit is influenced by various factors, of which the procedural technique followed by the dentist is an at most factor to be taken into account.[12] A technique will be considered to be completely successful only if it has a positive impact on the child's behavior. A lot of scales for assessing the behavior have been declared, of which Frankl behavior rating scale is considered to be the most reliable tool for assessing the behavior of the children in a dental setup.[13]

Dental anxiety and dental pain are two other important factors that influence the behavior of the child in a dental sitting. In a pediatric dental practice, management of dental fear and anxiety which influences the behavior of the child for the dental procedure is an at most factor to be taken into consideration. It has been well-documented that the difficulties associated with behavior management are not only related to the technical procedures involved but also to the emotional constituents such as the fear and anxiety.[14] Anxiety is an unpleasant emotional state arising without any objective source of danger. Anxiety in children during a dental procedure will not only have an impact on the child's dental attitude but will also affect the quality of the treatment.

Pain is another factor that has an impact on the child's behavior for the dental procedure. Pain is defined as an unpleasant sensory or emotional experience associated with actual or potential tissue damage. In pediatric patients, effective pain management is the cornerstone for successful behavior management.[15]

Various studies have been conducted using different rotary instrumentation for canal preparation in primary teeth and have concluded that there is a significant reduction in the chair side time which indirectly improves the cooperation of the child for the dental treatment.[8,16,17] However the direct association of the use of rotary instrumentation on the child's behavior and the anxiety levels has not been assessed. Determining the behavior of the children, their anxiety level, and the pain experienced by them on using rotary instruments is elemental to completely accept the use of NiTi files in routine pediatric dental practice. Hence, the aim of the present study was to compare and evaluate the child's behavior, anxiety level and the pain perception on using Manual and Rotary instrumentation for cleaning and debridement of the primary root canals.

Materials and Methodology

Study Design and Ethical Approval

The study was a single-blinded randomized controlled trial conducted at Saveetha Dental College and Hospitals, Chennai, Tamil Nadu, India. Ethical approval was obtained from the institutional ethical committee of Saveetha Institute of Medical and Technical Sciences prior to the start of the study (SRB/SDMDS15PED12) and was conducted in accordance with the Declaration of Helsinki.

The target population was children attending Saveetha Dental College for routine dental treatment. Children between the ages of 4 and 7 years showing cooperative behavior for the dental treatment in their first dental visit were invited to participate in the study. Children lacking cooperative ability, children with physical and psychological disabilities and with underlying systemic diseases requiring special health care needs were not invited to participate in the study. Also, children requiring pulpectomy in an abscessed or nonvital tooth, teeth showing signs of internal/pathological resorption, teeth with inadequate coronal tooth structure to receive Stainless Steel Crown and parents who were not willing and those who refused to sign the informed consent were also excluded from the study.

The participants' parents were explained about the treatment required for their children and the participation of their children into the study was voluntary. Informed consent was obtained from the parents of all the participants prior to the commencement of the treatment.

Sample Size Calculation

Sample size was calculated from a pilot study that was conducted with a total of 20 children (10 per group) meeting the above said inclusion and exclusion criteria. Sample size was calculated at 95% power with alpha error set 5% using G power and arrived to a total sample size of 88 (44 per group).

Randomization and Allocation Concealment

The participants were randomly allocated into two groups using a computer-generated randomization sequence. The sequence was numbered in advance and was sealed in an envelope to ensure adequate concealment. As the participant was recruited into the trial, the investigator opened the sequentially arranged envelopes to determine the type of instrumentation that has to be used for pulpectomy in that particular participant.

Study Procedure

Pulpectomies for all the included teeth were carried out by a single pediatric dentist in single visit. Topical local anesthetic agent (Precaine B, Pascal International, USA) was applied and the teeth were anesthetized with local anesthetic solution containing 2% lignocaine with 1:200,000 adrenaline (LOX* 2% ADRENALINE, Neon Laboratories limited, India). Rubber dam isolation was done (GDC Marketing, India). The superficial caries was removed using high speed sterile No. 330 round carbide burs (Mani, Inc., Japan). A standard access opening and coronal pulp tissue removal was done. The overlying dentinal shelf was reduced until all the canals were completely visible. The vitality of the tooth was confirmed by the presence of hemorrhage in the canal upon de-roofing. No.10 size K file (Mani, Inc., Japan) was used to explore and determine the patency of the canal. Working length determination was done with the help of pre-operative radiograph and was kept 1 mm short of the apex.

In group I, the canals were prepared using K files (Mani, Inc., Japan) till size 30 in quarter pull turn method. The canals were irrigated with saline between each file size.

In group II, a simplified protocol for the use of ProTaper rotary files (Dentsply India Pvt. Ltd, India) in primary teeth were followed as suggested by Kuo C et al.[4] SX and S2 ProTaperfiles were used for the canal preparation of the primary teeth sequentially using an X-Smart motor (Dentsply India Pvt. Ltd., Delhi, India) at a rotational speed of 300 r.p.m. with torque set at the lowest level.

SX file was inserted into the canals in bucco-lingual brushing motion to remove the over-lying dentin and gain straight line access. This was followed by insertion of the S2 file till the determined working length. On removal of the file from the canal, the pulp tissue was commonly seen wrapped to it. The primary root canals were thoroughly irrigated using saline. The files were discarded after use in five teeth and no attempt was made to force the files beyond the point of resistance if felt to prevent the fracture of the instrument within the canal.

During canal preparation, the operator assessed the behavior of the child using Frankl's behavior rating scale. The criteria used to categorize the participants as definitely positive, positive, negative, and definitely negative. The anxiety level of the children on instrumentation was recorded using Venham's Interval Rating Scale. This scale scores from 0 to 5 describing the anxiety level of the children in the increasing order. Behavior and anxiety scores were recorded by the dentist who performed the treatment. Tell Show and Do technique and use of euphemisms were commonly employed for management of the children of both the groups. In cases where the child showed negative behavior voice control, HOME, and physical restraints were used for managing the children.

After canal preparation, the child was assessed for the pain experienced during instrumentation using Wong Baker FACES Pain Scale (Fig. 1). This self- reported scale was explained to the children and they were asked to point out the face that depicts their pain level during the canal instrumentation.

The canals were then dried thoroughly using sterile paper points. A premixed paste of calcium hydroxide and iodoform was used to obturate the root canals (Metapex, Meta Biomed Co., Ltd., Korea). The obturated canals were given Glass Ionomer Cement entrance filling (Shofu, Shofuinc. Japan) and was restored with Stainless Steel crowns (3M ESPE).

S
T 7.0
( )5.
I ıre
t nd
t ely.
M ety
a os.
S ıip
k in.

I
E es,
p ıts
v he
F nd
r rst

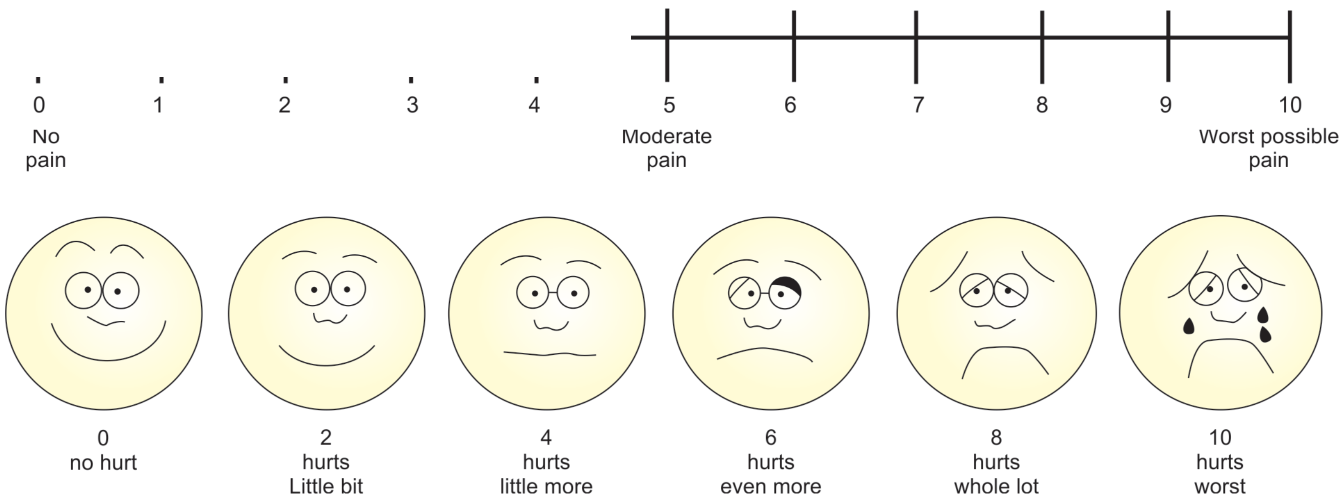

Fig. 1: Wong Bakers FACES pain scale

Conclusion

The results of the present study draw to the following conclusions:

- The behavior of the children during canal preparation with rotary files is more positive when compared to those children in whom manual files were used.
- The anxiety level of the children was comparatively lesser on instrumenting with rotary files when compared to the manual files.
- The intensity of pain experienced by the children during canal preparation with rotary files was much lower than manual instrumentation.
- It is preferable to use rotary instrumentation for primary canal preparation in pediatric practice as it has a more positive influence on the behavior of the children, which eventually determines the success of the treatment.

References

1. Ranly DM, Garcia-Godoy F. Current and potential pulp therapies for primary and young permanent teeth. J Dent 2000;28:153–161. DOI: 10.1016/s0300-5712(99)00065-2
2. Fuks AB. Vital pulp therapy with new materials for primary teeth: new directions and treatment perspectives. Pediatr Dent 2008;30:211–219. PMID: 18615986.
3. Haapasalo M, Endal U, Zandi H, et al. Eradication of endodontic infection by instrumentation and irrigation solutions. Endodontic Topics 2005;10:77–102. DOI: 10.1111/j.1601-1546.2005.00135.x
4. Kuo C, Wang Y, Chang H, et al. Application of Ni-Ti rotary files for pulpectomy in primary molars. J Dent Sci 2006;1:10–15. DOI: 10.30086/JDS.200603.0002
5. Silva LAB, Leonardo MR, Nelson-Filho P, et al. Comparison of rotary and manual instrumentation techniques on cleaning capacity and instrumentation time in deciduous molars. J Dent Child 2004;71:45–47. PMID: 15272656.
6. Barr ES, Kleier DJ, Barr NV. Use of nickel-titanium rotary files for root canal preparation in primary teeth. Pediatr Dent 2000;22:77–78. PMID: 10730297.
7. Rodd HD, Waterhouse PJ, Fuks AB, et al. British Society of Paediatric Dentistry. Pulp therapy for primary molars. Int J Paediatr Dent 2006;16:15–23. DOI: 10.1111/j.1365-263X.2006.00774.x
8. Crespo S, Cortes O, Garcia C, et al. Comparison between rotary and manual instrumentation in primary teeth. J Clin Pediatr Dent 2008;32:295–298. DOI: 10.17796/jcpd.32.4.l57l36355u606576
9. Pinkham JR. Behavioral themes in dentistry for children: 1968–1990. ASDC J Dent Child 1990;57:38–45. PMID: 2136747.
10. Roberts JF, Curzon MEJ, Koch G, et al. Review: behaviour management techniques in paediatric dentistry. Eur Arch Paediatr Dent 2010;11:166–174. DOI: 10.1007/BF03262738
11. Shinohara S, Nomura Y, Shingyouchi K, et al. Structural relationship of child behavior and its evaluation during dental treatment. J Oral Sci 2005;47:91–96. DOI: 10.2334/josnusd.47.91
12. McDonald RE, Avery DR, Dean AJ. In: Dentistry for the child and adolescent. 8th edition. St. Louis: Mosby, 2004; p. 729.
13. Frankl S, Shiere F, Fogels H. Should the parent remain with the child in the dental operatory J Dent Child 1962; 29:150–163.
14. Brill W. The effect of restorative treatment on children's behavior at the first recall visit in a private pediatric dental practice. J Clin Pediatr Dent 2002;26:389–393. DOI: 10.17796/jcpd.26.4.r1543673rx055355
15. Ashkenazi M, Blumer S, Eli I. Effectiveness of computerized delivery of intrasulcular anesthetic in primary molars. J Am Dent Assoc 2005;136:1418–1425. DOI: 10.14219/jada.archive.2005.0056
16. Ochoa-Romero T, Mendez-Gonzalez V, Flores-Reyes H, et al. Comparison between rotary and manual techniques on duration of instrumentation and obturation times in primary teeth. J Clin Pediatr Dent 2011;35:359–363. DOI: 10.17796/jcpd.35.4.8k013k21t39245n8
17. Makarem A, Ravandeh N, Ebrahimi M. Radiographic assessment and chair time of rotary instruments in the pulpectomy of primary second molar teeth: a randomized controlled clinical trial. J Dent Res Dent Clin Dent Prospects 2014;8:84–89. DOI: 10.5681/joddd.2014.015
18. Sweet CA. Procedure for treatment of exposed and pulpless deciduous teeth**Read before the Section on Mouth Hygiene and Preventive Dentistry at the Seventy-First Annual Session of the American Dental Association, Washington, DC, Oct. 9, 1929. J Am Dent Assoc 1930;17:1150–1153.
19. Coll JA, Sadrian R. Predicting pulpectomy success and its relationship to exfoliation and succedaneous dentition. Pediatr Dent 1996;18:57–63. PMID: 8668572.
20. Rathnam A, Madan N, Madan N. The language of pain: A short study. Contemp Clin Dent 2010;1:142–145. DOI: 10.4103/0976-237X.72778
21. Ramezanali F, Afkhami F, Soleimani A, et al. Comparison of cleaning efficacy and instrumentation time in primary molars: Mtwo rotary instruments vs. Hand K-files. Iran Endod J 2015;10:240–243. DOI: 10.7508/iej.2015.04.006
22. Govindaraju L, Jeevanandan G, Subramanian EMG. Comparison of quality of obturation and instrumentation time using hand files and two rotary file systems in primary molars: a single-blinded randomized controlled trial. Eur J Dent 2017;11:376–379. DOI: 10.4103/ejd.ejd_345_16
23. Pinheiro SL, Araujo G, Bincelli I, et al. Evaluation of cleaning capacity and instrumentation time of manual, hybrid and rotary instrumentation techniques in primary molars. Int Endod J 2012;45:379–385. DOI: 10.1111/j.1365-2591.2011.01987.x
24. Azar MR, Safi L, Nikaein A. Comparison of the cleaning capacity of Mtwo and Pro Taper rotary systems and manual instruments in primary teeth. Dent Res J (Isfahan) 2012;9:146–151. DOI: 10.4103/1735-3327.95227
25. Bharuka SB, Mandroli PS. Single- versus two-visit pulpectomy treatment in primary teeth with apical periodontitis: A double-blind, parallel group, randomized controlled trial. J Indian Soc Pedod Prev Dent 2016;34(4):383–90. DOI: 10.4103/0970-4388.191429
26. Brill WA. Child behavior in a private pediatric dental practice associated with types of visits, age and socio-economic factors. J Clin Pediatr Dent 2000;25:1–7. DOI: 10.17796/jcpd.25.1.545025p1g72x730q
27. Gustafsson A, Arnrup K, Broberg AG, et al. Psychosocial concomitants to dental fear and behaviour management problems. Int J Paediatr Dent 2007;17:449–459. DOI: 10.1111/j.1365-263X.2007.00883.x